

The more, the merrier? Establishing a dose-response relationship for the effects of cognitive control training on depressive symptomatology

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Abstract

Cognitive impairments are frequently observed after remission from depression. There is evidence to show that Cognitive Control Training (CCT) can reduce such impairments and related risk for recurrent episodes. However, it is currently unclear how many CCT sessions are required for short and long-term effects. This randomized controlled trial (RCT) investigates the dose-response relationship of CCT in a sample of individuals with remitted depression (RMD; n=216). Participants were randomly assigned to one of six arms, each receiving a different amount of training sessions (0, 1, 5, 10, 15 or 20 sessions). We found task-specific, but no near cognitive transfer. Mixed-effects models showed that a minimum of 10 training sessions was required for a significant decrease in depressive symptoms shortly after training, but this was not maintained at three- or six-month follow-up. We observed reductions in perseverative thinking in all groups receiving 10 or more CCT sessions, which remained present until six-month follow-up.

Keywords: Cognitive control training · Depression · Recurrence of depression · Dose-response · Working memory training

Introduction

Depression is a widespread health concern, affecting millions of people around the world (World Health Organization, 2024). The prevalence of Major Depressive Disorder (MDD) is known to be especially high in low and low-to-middle income countries (Global Burden of Disease Collaborative Network, 2019), indicating the need for cost-effective and scalable interventions. Current treatments targeting depressive symptomatology have proven moderately effective in the acute phase, but are less effective in preventing recurrence of depression, with estimated recurrence rates between 40% to 60% (Bockting et al., 2015).

One of the core characteristics of MDD is impaired cognitive functioning (Culpepper et al., 2017), which often remains present after recovery (Schmid & Hammar, 2021; Semkovska et al., 2019). Although initially considered a byproduct of depressed mood, more recently cognitive impairments have been identified as a key factor in the recurrence of depression (Buckman et al., 2018; Jaeger et al., 2006). Therefore, interventions have been developed to target these cognitive impairments specifically, such as Cognitive Control Training (CCT; see Siegle et al., 2007; Koster et al., 2017). CCT often uses computerized tasks to train cognitive control, the ability to flexibly regulate thoughts, emotions and behavior in function of one's goals (Cohen, 2017). Considering the important role of cognitive control in emotion-regulation processes (Joormann & D'Avanzato, 2010; Joormann & Vanderlind, 2014; Mor & Daches, 2015), improving cognitive control through CCT could lead to decreased risk for recurrence of depression. Previous studies examining CCT have found positive effects on depressive symptomatology immediately after training (Siegle et al., 2007; Calkins et al., 2015), as well as long-term benefits such as decreased need for healthcare services (Siegle et al., 2014) or decreased recurrence of depression (Hoorelbeke et al., 2021). Moreover, recent systematic reviews and meta-analyses have reported significant small to medium effects of CCT on depressive symptomatology (Motter et al., 2016; Woolf et al., 2021).

CCT can be operationalized using different cognitive tasks. In prior research, some studies have focused on using one task (e.g., dual n-back training; Beloe & Derakshan, 2019), whilst others have employed a combination of different tasks (e.g., Cogmed; Lengvenyte et al., 2019). The advantage of using only one task for CCT is that task heterogeneity is low and effects can be attributed to the specific CCT operationalization, whereas using multiple tasks may potentially increase treatment adherence due to less repetitive training. One single-task operationalization which has shown promising results in earlier CCT research in MDD (Siegle et al., 2007), is an adaptive version of the Paced Auditory Serial Addition Task (PASAT;

Gronwall, 1977). A recent meta-analysis examining the effectiveness of an adaptive PASAT (aPASAT) training has found small to moderate, significant effects on depressive symptomatology and rumination, both immediately after training and at long-term follow-up (Vander Zwahlen et al., 2023). During the PASAT, participants are presented with a stream of digits and are asked to continuously make the sum of the last two heard digits. Where the standardized PASAT typically presents digits at fixed speeds (e.g., 3000ms, 2000ms and 1500ms), the speed of presentation of the auditory digits in the aPASAT changes, depending on the performance of an individual. This is often done by increasing speed by 100ms for every four consecutive correct answers and decreasing speed by 100ms for every four consecutive incorrect answers.

In previous CCT studies, the aPASAT has been used with a different number of sessions (e.g., three sessions: Calkins et al., 2015; five sessions: Segrave et al., 2014; ten sessions: Hoorelbeke et al., 2017, 2021; twelve sessions: Sommer & Plewnia, 2021), but it is currently unclear how many sessions are required to find beneficial effects on cognitive functioning or depressive symptomatology. Here, it is important to know what a *minimal dose* is to obtain clinical effects and to identify *optimal dosage*. If the number of sessions is too low, the required effects might occur insufficiently or be short-lived, but if the number of sessions is too high, unwanted side-effects could occur (e.g., attrition or decreased motivation). A large number of sessions might be difficult to incorporate for individuals with (partially) remitted depressive symptoms and could lead to low uptake, fatigue or drop-out.

Some initial research examining dose-response effects of training regimes for CCT has been conducted. For instance, one interesting study examined dose-response functions of a commercial cognitive training in a very large dataset ($n=107.000$; Osman et al., 2023). A non-randomized sample took part in an online battery of cognitive assessments before and after 10 weeks of cognitive training. Strikingly, analyses of dose-response functions allowed for the separation of practice effects due to repeated testing and transfer effects. Where direct practice effects diminished with age, transfer effects remained constant, indicating that these processes do not involve identical learning mechanisms (Osman et al., 2023). In another study, Long et al. (2023) used a dual n-back training in a student sample and compared 7-day and 15-day training procedures. Where the 7-day training resulted in a small reduction of emotional responses compared to an active control condition, the 15-day training improvement was more pronounced, suggesting a dose-response effect. Similarly, the meta-analysis of Launder et al. (2021) found a moderating effect of dose on overall cognition, with larger CCT doses being

associated with greater effect size estimates. However, this was not statistically significant for depressive symptomatology. A meta-regression from Vander Zwalm et al. (2023) also found no significant dose-response relationship on depressive symptomatology following aPASAT training. However, this is likely due to low variability in aPASAT dosage used in prior studies. Currently, it is not clear if an optimal CCT dosage exists and how its effects unfold over time (Vander Zwalm et al., 2022). In a meta-analysis by Traut et al. (2021), it has been posited that perhaps sufficient cognitive capacity is required in order to efficiently train cognitive control, which could mean that improvements following CCT could also be non-linear for some individuals. Moreover, Traut et al. (2021) indicate that most cognitive gains are possible in individuals with higher cognitive impairments.

The importance of examining dosage and timing of interventions in relation to depression has become clear from research on other well-established treatments for depression. For antidepressant medication, research has indicated that improvements usually occur after several weeks (Machado-Vieira et al., 2010) and NICE guidelines propose a continued treatment of 6 months after remission (NICE, 2022). In cognitive behavioral therapy, the occurrence of sudden gains is well known, often showing a sudden decrease in depressive symptomatology after 4 sessions (Tang et al., 2005), suggesting that improvements may also be non-linear. In context of depression prevention after remission, it could be of importance to account for similar possibilities in studying effects of CCT.

In terms of investigating the temporal unfolding of CCT effects, a clear understanding of *how* CCT affects depressive symptomatology is currently lacking. One possible avenue is through the effects of CCT on rumination. Rumination and repetitive negative thinking have been closely associated with, and are an important symptom of, MDD (Nolen-Hoeksema et al., 2008). It has been posited that improved cognitive functioning through CCT could influence decoupling from ruminative thinking, reducing depression risk (Vander Zwalm et al., 2022). In line with this, a previous randomized controlled trial (RCT) found cognitive control gains to be related to a reduction in depressive complaints over time, which was partially mediated by immediate improvements in ruminative thinking following training (Hoorelbeke & Koster, 2017). However, evidence in this line of thinking is inconsistent, with some studies finding positive effects of CCT on rumination (Hoorelbeke & Koster, 2017; Siegle et al., 2014; Vanderhasselt et al., 2015), where others found no effects (Ferrari et al., 2021; Lass et al., 2021; Vervaeke et al., 2020). Another possible path of how CCT could work is proposed by the capacity-efficiency model, which focusses on examining transfer effects (for review, see von

Bastian et al., 2022). Several types are often distinguished: task-specific transfer (i.e., improvements on a standardized version of the trained task), near transfer (i.e., improvements on untrained tasks for which the same cognitive construct is required, e.g. working memory) and far transfer (i.e., improvements on other cognitive and affective constructs). In this capacity-efficiency framework, CCT could increase cognitive capacity, cognitive efficiency, or both. Given that little evidence has suggested increased cognitive *capacity* following cognitive training, but substantial evidence has supported training-induced improvements in cognitive *efficiency*, more insight into underlying mechanisms of training-induced cognitive change is required (von Bastian et al., 2022). In order to examine transfer effects, it has been recommended that CCT studies include outcomes on different transfer levels (see Koster et al., 2017).

To our knowledge, no previous research has conducted an experimental manipulation of aPASAT training dosage to examine its effects on depressive symptoms, which is required for clinical implementation. In this RCT, individuals in remission from depression (RMD) were randomized across 6 groups, receiving either 0, 1, 5, 10, 15 or 20 aPASAT training sessions (15 minutes each), going beyond aPASAT dosages typically used in the literature (Vander Zwalm et al., 2023). Cognitive and emotional transfer effects were measured shortly after training, as well as at a three- and six-month follow-up.

Transparency and Openness

This paper presents the findings of an open label randomized controlled trial (RCT) that was preregistered on ClinicalTrials.Gov with identifier NCT05166798. The study was approved by the medical ethical comity of Ghent University Hospital (BC-10551) and written informed consent was obtained for all participants. All data and code are available on Open Science Framework (<https://osf.io/k4paq>). We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study.

Method

The study procedure consisted of (a) a telephone screening; (b) an in-person screening and baseline measure in the lab; (c) a training phase completed at home; (d) a post measure in the lab; (e) an online three- and (f) six-month follow-up completed at home. In accordance with an a priori power analysis (based on a medium effect size from a similar trial, Hoorelbeke & Koster, 2017) and aiming to achieve statistical power of 80%, a total sample of 216 participants were recruited. Code for the power analysis can be found on OSF (<https://osf.io/k4paq>). After screening for eligibility, participants were randomized across six groups and received either 0 (waitlist control group), 1, 5, 10, 15 or 20 CCT sessions.

Participants

Individuals with remitted depression (RMD) were recruited through a combination of classical (printed advertisements, local newspapers) and social media. Inclusion criteria were a history of ≥ 1 depressive episode(s), currently being in remission (≥ 3 months) and access to a computer with an internet connection. Exclusion criteria were (a) an ongoing depressive episode; (b) a current or previous psychotic disorder; (c) current or previous neurological impairments; and (d) excessive substance abuse. The use of antidepressant medication was allowed if kept at a constant level. Eligibility was first assessed using a brief telephone screening during which the relevant modules of the Mini-International Neuropsychiatric Interview (MINI; Sheehan et al., 1998; Van Vliet & De Beurs, 2007) were administered to assess current and previous depressive symptomatology based on criteria of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5; American Psychiatric Association, 2013). Qualifying participants were invited to Ghent University for a second screening, where licensed clinical psychologists repeated the screening procedure and checked the inclusion criteria.

Training

For CCT, we relied on a computerized gamified aPASAT training which was developed based on input of RMD individuals to optimize user engagement (Vervaeke et al., 2018). Among others, gamification elements included feedback on training task progress, as well as the use of vivid colors (for a detailed description, see Vervaeke et al., 2019). To tailor the aPASAT training to the individual, the starting interstimulus interval (ISI) of the first training session was determined by the performance on the standardized PASAT. The standardized PASAT has three blocks of 60 trials, at 3000ms, 2000ms and 1500ms. If a participant scored less than 51% accuracy in the 3000ms block, the starting ISI of the first training session was 4000ms. For participants reaching $\geq 51\%$ accuracy on the first block, while obtaining less than

51% accuracy in the 2000ms block, the ISI for the first training sessions was set at 3500ms. Similarly, for participants scoring higher on the previous blocks, while reaching less than 51% accuracy on the 1500ms block, an ISI of 2500ms was used. Finally, if a participant scored higher than 51% accuracy on all blocks of the standardized PASAT, the initial ISI of the first training session was set to 2000ms. To ease participants at the start of each new training session, the starting ISI of subsequent training sessions was determined by the median ISI of the previous session, augmented with 500ms. This is in line with prior CCT studies in which the gamified aPASAT was used (e.g., Vervaeke et al., 2021) and allows for gradual improvement over the course of training sessions.

Outcome measures

Outcome measures consisted of self-report questionnaires and cognitive tasks. We included three measures of cognitive transfer. *Task-specific transfer* was assessed with a standardized version of the PASAT, using three blocks of 60 trials each, where the interstimulus interval (ISI) increased with each block, using an ISI of 3000ms, 2000ms and 1500ms for each block respectively. Task-specific transfer was assessed using the mean accuracy over the three blocks. *Near transfer* was measured with a dual n-back task (Jaeggi et al., 2008), which presented a visual (blue square) and auditory stimulus (letter) simultaneously. Three levels were used (1-back, 2-back and 3-back), each level consisting of three blocks, where each block contained $20 + n$ trials. The measure of interest was the proportion of hits minus false alarms averaged over all n-back levels (cf., Jaeggi et al., 2010). The dual n-back task was only assessed during lab visits (baseline and post-training). In addition, we included an indicator for *subjective cognitive functioning*, for which we relied on the Effortful control (EC) subscale of the Adult Temperament Questionnaire (ATQ-EC; Hartman & Rothbart, 2001; Rothbart et al., 2000). We found good internal consistency for the ATQ-EC (Cronbach's $\alpha = 0.81$).

The primary outcome, *depressive symptomatology*, was assessed with the Beck Depression Inventory II (BDI-II; range: 0–63; Beck et al., 1996; Van der Does, 2002), which showed excellent internal consistency (Cronbach's $\alpha = 0.90$). In addition, to examine effects of CCT on a broad range of *residual depressive complaints*, the Remission from Depression Questionnaire (RDQ; range 0-82; Zimmerman et al., 2013a, 2013b) was used, which also showed excellent internal consistency (Cronbach's $\alpha = 0.96$). After reverse scoring items of the positive mental health, coping ability, functioning, life satisfaction, and general sense of well-being subscales, the total score of the RDQ was obtained by summing up item scores, with lower total scores reflecting better functioning. The degree of *burnout complaints* was measured

with the short version of the Burnout Assessment Tool (BAT; range: 12-60; Schaufeli et al., 2019, 2020). Aggregating the working and non-working versions of the BAT, we found good internal consistency (Cronbach's $\alpha = 0.89$). *Repetitive negative thinking* was assessed with the Perseverative Thinking Questionnaire (PTQ; range: 0-60; Ehring et al., 2011; Ehring et al., 2012), which showed excellent internal consistency (Cronbach's $\alpha = 0.94$). Broader use of (mal)adaptive emotion regulation strategies was measured with the Cognitive Emotion Regulation Questionnaire (CERQ; Garnefski et al., 2001). Subscales for adaptive strategies include acceptance, taking perspective, planning, positive reappraisal, and positive reflection. Maladaptive strategies assess the use of catastrophizing, ruminating, blaming yourself and blaming others. Compound scores were calculated (range adaptive compound score: 20-100, range maladaptive compound score: 16-80). Internal consistency was high for the adaptive score (Cronbach's $\alpha = 0.88$) and adequate for the maladaptive score (Cronbach's $\alpha = 0.79$).

Inclusion of the *Credibility and Expectancy Questionnaire* (CEQ; Devilly & Borkovec, 2000) allowed to investigate how the participant experienced the training procedure and whether their perspective on CCT changed over time. CEQ subscales were created in line with Devilly & Borkovec (2002): after standardization using Z-scores, sum scores were created from the first three items to create the credibility subscale and the last three items were summed to create the expectancy subscale. The CEQ displayed adequate consistency (Cronbach's $\alpha = 0.73$). Lastly, the possible influence of *stressful life events* was evaluated using the List of Threatening Experiences (LTE; Brugha & Cragg, 1990; Rosmalen et al., 2012). All questionnaires were completed at all timepoints, except for the CEQ which was only assessed at baseline and post, and the LTE, which was only assessed at three- (FU1) and six-month (FU2) follow-up.

Procedure

All participants received standardized psychoeducation through the cognitive training platform and were randomized across groups, after which they filled in baseline questionnaires and completed two cognitive tasks: a non-adaptive PASAT and a dual n-back task were used to assess baseline level of cognitive functioning. After baseline, the waitlist control group was informed that they did not have to complete training sessions at home. The experimental groups were asked to complete one training session of 15 minutes per day and were instructed to complete their total required number of training sessions (i.e., 1, 5, 10, 15 or 20). After each session, participants scheduled the next session and automated reminders were sent via email. All groups were invited back to the lab four weeks after baseline for the post measurement,

which consisted of the same questionnaires and cognitive tasks as during the baseline measurement. Participants were invited through email to complete the 3 and 6 month follow-ups using LimeSurvey (LimeSurvey, 2024) for the self-report measures and the Cogtraining platform (<https://cogtrain2.ugent.be>) for the non-adaptive PASAT. After the six-month follow-up, participants were reimbursed €40 and were offered a new account on the training platform which allowed them to continue the training after completing participation to the study.

Data analysis

Analyses were conducted in R, version 4.3.2 (R Core Team, 2023). A list of the used R packages and their version numbers can be found in the supplementary material (table A1). Linear mixed models were constructed with group and time as fixed effects, Group x Time as interaction effect and participant as random intercept:

$$outcome \sim group + time + group * time + (1 | participant)$$

For examination of outcomes with only two timepoints (dual n-back, CEQ), two models were fitted: one without and one with the group and time interaction. Likelihood ratio tests (LRT; Glover & Dixon, 2004) and Akaike Information Criterion (AIC; Akaike, 1974) were used to assess model fit. LRT assess whether adding specific terms significantly improves model fit and AIC provides a quantitative measure to compare models, accounting for both goodness of fit and model complexity. If the likelihood ratio test was not significant (i.e., there was no significant difference between the models), the less complex model (i.e., the model without the group and time interaction) was chosen. Similarly, when the difference in AIC was larger than 2 between two models, the model with lower AIC was considered the preferred model. For analysis of the outcomes with four timepoints where the homoscedasticity assumption was violated, Poisson and negative binomial regression models were compared and the best fitted model was chosen based on (lower) AIC, indicating better model fit. Contrast testing was done using the multcomp package in R (Piepho, 2004) and the familywise error rate was controlled for using single-step method. All data was analyzed using intention-to-treat (ITT) analyses. To gain insight in the effect size for CCT effects on the primary outcome (i.e., depressive symptomatology, assessed with the BDI-II), post-means Hedges' g effect sizes were

calculated¹ and converted into Number Needed to Treat (NNT) using the Kraemer and Kupfer method (Kraemer & Kupfer, 2006) from the R package dmetar (Harrer et al., 2019).

Due to technical difficulties, some PASAT (n=3; 1.4% of total sample) and dual n-back (n=1; 0.5% of total sample) measurements at baseline were lost. Two PASAT measurements at FU1 were lost due to technical issues with participant's personal computers (1.0% of FU1 PASAT data). One PASAT measurement was deleted at FU2 due to a participant only completing the task partially and one more PASAT measurement at FU2 was lost due to technical issues with the personal computer of a participant (1.0% of FU2 PASAT data).

¹ We choose Hedges' g instead of Cohen's d effect sizes to account for the sample size of the individual groups. We opted for post-means effect sizes, because the main research question requires between-group comparisons. For the interested reader, pre-post (within-group) Hedges' g effect sizes are available in the supplementary materials (table A2).

Results

Group characteristics

After the screening phase, a total of $n = 216$ participants were randomized across groups (CONSORT flow diagram, see Fig.1). Descriptive information can be found in Table 1. The first participant had their baseline assessment on February 2nd 2022, and data collection was completed by April 17th 2024. The mean number of days between baseline and post assessments ($M (SD) = 29.4 (2.5)$), baseline and FU1 ($M (SD) = 88.8 (4.8)$) and baseline and FU2 ($M (SD) = 177.9 (13.8)$) did not statistically differ between groups (all p 's $> .451$). The six groups did not significantly differ on demographic variables (see table 1). Furthermore, no significant group differences were found on baseline depressive symptomatology ($F(5, 210) = 1.63, p = .154$), credibility ($F(5, 210) = 1.18, p = .320$), expectancy ($F(5, 210) = 1.56, p = .172$), or on cognitive task performance (PASAT: $F(5, 207) = 0.26, p = .936$; dual n-back: $F(5, 209) = 0.36, p = .875$). The number of significant stressful life events did not differ between groups at three-month follow-up (LTE: $F(5, 197) = 1.12, p = .353$), nor at six-month follow-up (LTE: $F(5, 194) = 1.61, p = .159$). Finally, a Wilcoxon rank sum test found that participants who dropped out during the study ($n = 16$) had significantly lower baseline PASAT accuracy scores compared to participants who completed the study ($W = 2265.5, p = 0.014$).

Table 1. Group Characteristics

Characteristic	GR 0 (N=37)	GR 1 (N=38)	GR 2 (N=37)	GR 3 (N=36)	GR 4 (N=32)	GR 5 (N=36)	Statistic
Age (<i>SD</i>)	36.0 (13.4)	39.6 (16.6)	38.4 (13.8)	40.1 (12.2)	35.9 (11.1)	39.9 (12.3)	$F(5, 210) = 0.738, p = .596$
Sex (male:female)	7:30	13:25	8:29	11:25	14:18	11:25	$\chi^2(5) = 6.65, p = .248$
Gender (male:female:other)	7:28:2	13:22:3	8:29:0	11:25:0	14:18:0	10:25:1	Fisher's exact test: $p = 0.173$
Highest degree obtained							Fisher's exact test: $p = 0.190$
Primary school	1	0	0	0	0	1	
Secondary school	14	18	13	11	8	9	
Bachelor's degree	12	13	13	8	13	14	
Master's degree	7	7	10	17	11	11	
PhD	3	0	1	0	0	1	
Marital status							Fisher's exact test: $p = 0.870$
Single	23	18	16	16	17	19	
Living together	5	2	6	4	3	3	
Married	8	15	11	11	10	11	
Divorced	1	3	3	5	2	3	
Widower	0	0	1	0	0	0	
Ethnicity							Fisher's exact test: $p = 0.677$
Caucasian	36	34	35	35	29	36	
North-African	0	1	0	0	0	0	
Asian	0	1	0	0	2	0	
Latin-American	0	1	1	0	1	0	
Other	1	1	1	1	0	0	
Age of first depressive episode (<i>SD</i>)	24.8 (11.3)	26.2 (12.9)	23.2 (12.4)	27.1 (13.0)	23.0 (9.6)	26.8 (12.9)	$F(5, 210) = 0.77, p = .575$
Number of past depressive episodes (<i>SD</i>)	4.2 (5.1)	3.2 (3.4)	4.5 (8.4)	3.3 (3.3)	4.0 (5.9)	3.4 (3.4)	$F(5, 210) = 0.37, p = .868$
Current ADM use (%)	40.5	36.8	51.4	47.2	46.9	55.6	$\chi^2(5) = 3.50, p = .624$
Current psychotherapy (%)	51.4	42.1	37.8	44.4	46.9	36.1	$\chi^2(5) = 2.39, p = .793$
Past suicide attempts (%)	16.2	18.4	29.7	22.2	18.8	25.0	$\chi^2(5) = 2.69, p = .747$
Past inpatient treatment (%)	16.2	28.9	35.1	41.7	28.1	33.3	$\chi^2(5) = 6.31, p = .277$

Note: ADM = antidepressant medication; GR 0 = waitlist control group; GR 1 = one CCT session group; GR 2 = five CCT sessions group; GR 3 = ten CCT sessions group; GR 4 = fifteen CCT sessions group; GR 5 = twenty CCT sessions group.

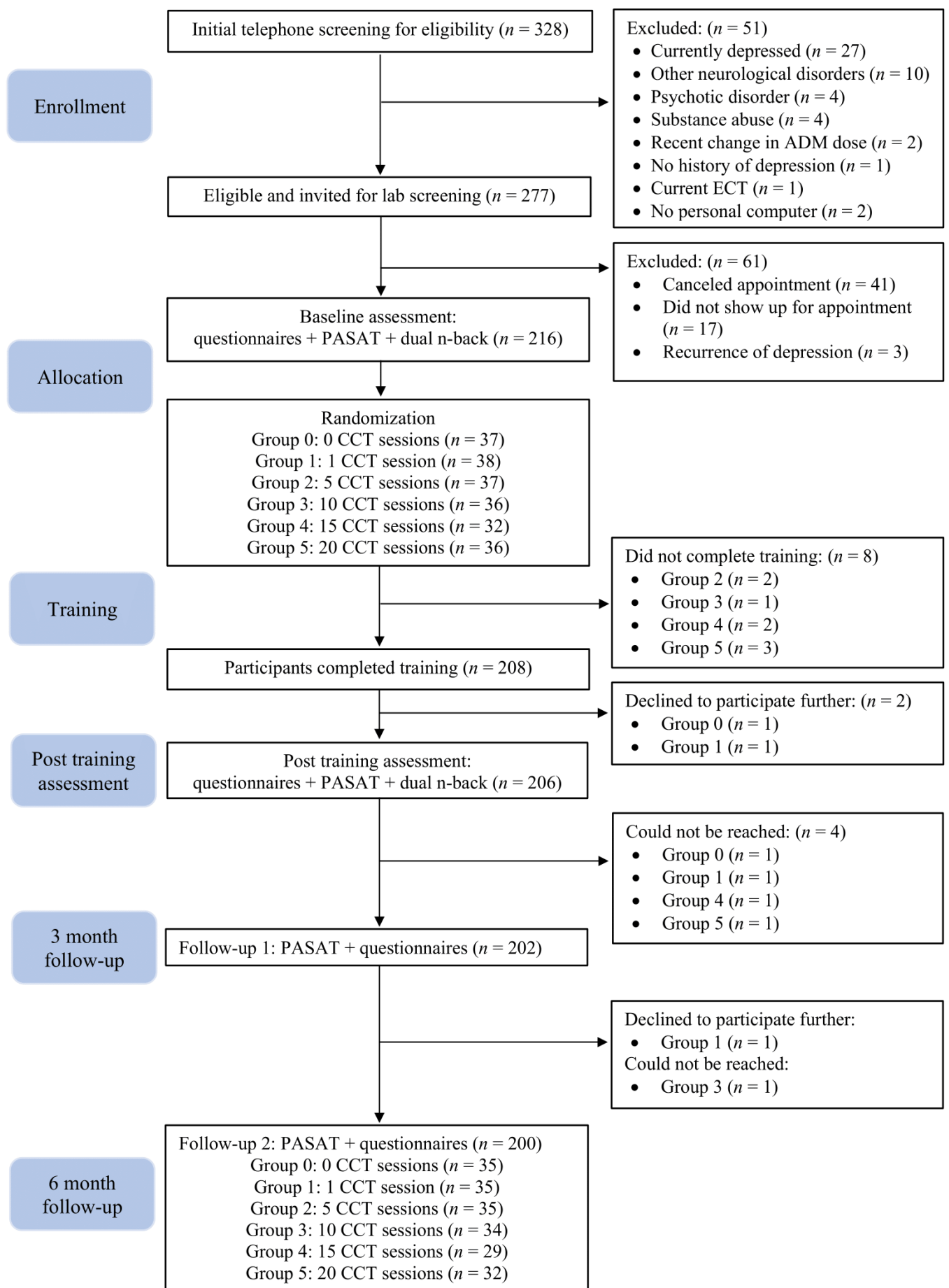


Fig. 1 CONSORT flowchart

Training progression and cognitive transfer

In line with recent aPASAT studies, the starting ISI of the first training session was determined by the participant's performance on the standardized PASAT. Across all groups, 41.6% of participants started at 4000ms (n=74), 53.4% at 3500ms (n=95), 4.5% at 2500ms (n=8), and 0.6% at 2000ms (n=1). There were no group differences in the initial ISI of the first training session ($p > .35$, Fisher's exact test). The median ISI of each aPASAT training session is often used to examine training progression over time (e.g., Hoorelbeke et al., 2015, Vervaeke et al., 2021). This typically follows a logarithmic trend: a strong initial decrease, followed by a plateau. Figure 2 shows the means of median ISI per session over time for all groups who performed aPASAT training. Linear regression analysis was conducted to examine the relationship between the group means of median ISI and the natural logarithm of session number ($\beta = -367.79$, $SE = 10.17$, $t = -36.16$, $p < .001$). The overall model was statistically significant ($F(1, 49) = 1308$, $p < .001$, $R^2_{adj} = 0.96$), indicating that as the session number increases, the ISI decreases across all groups (see supplementary material figure A1).

Task-specific transfer

The standardized PASAT was performed at all four time points. Similar to prior research, all groups saw a significant increase in performance from baseline to post and follow-up assessments. The linear mixed model found a significant Group x Time interaction effect ($\chi^2(15) = 627.29$, $p < .001$). After correcting for multiple testing, all groups at all time points showed significant increases, indicating all groups demonstrated higher accuracy scores at later time points, compared to baseline. As expected, the model predicted higher estimates for the higher dosage groups, demonstrating that individuals with more training sessions displayed higher accuracy on the standardized PASAT at post and both follow-up measures. Detailed results from the multiple comparisons analysis are reported in the supplementary materials, table A3 and plot A2.

Near transfer

Since the dual n-back was only performed at baseline and post-assessment, we examined if removing the group and time interaction from our model would improve model fit. We conducted a likelihood ratio test to compare the fit of two linear mixed-effects models. The first model included only the main effects of group and time, along with a random intercept for each participant. The second model included the main effects of group and time, along with their interaction, and a random intercept for each participant. The LRT indicated a significant

improvement in model fit with the inclusion of the interaction term ($\chi^2(6) = 42.44, p < .001$). When examining the two models on AIC, the second model also showed better fit (AIC model 1: 1137.6, AIC model 2: 1120.3). Therefore, we retained the model with the interaction term in our analyses. The mixed-effects model found a significant main effect of time ($\beta = 0.35, SE = 0.13, t(200.32) = 2.62, p = 0.009$), indicating a positive change over time. No main effect of group or interaction effect between time and group was found (all p 's > 0.52). This indicates that the dual n-back outcome variable was estimated 0.35 higher at post-measure compared to baseline, and no group differences emerged. Conditional R^2 (i.e., the variance explained by the entire model, both fixed and random effects) was 0.72.

After controlling for multiple testing, no significant changes in effortful control emerged from the ATQ-EC for any group at any time point (all p 's > 0.06). The model explained 69% of the conditional variance.

Means of median ISI per session over time

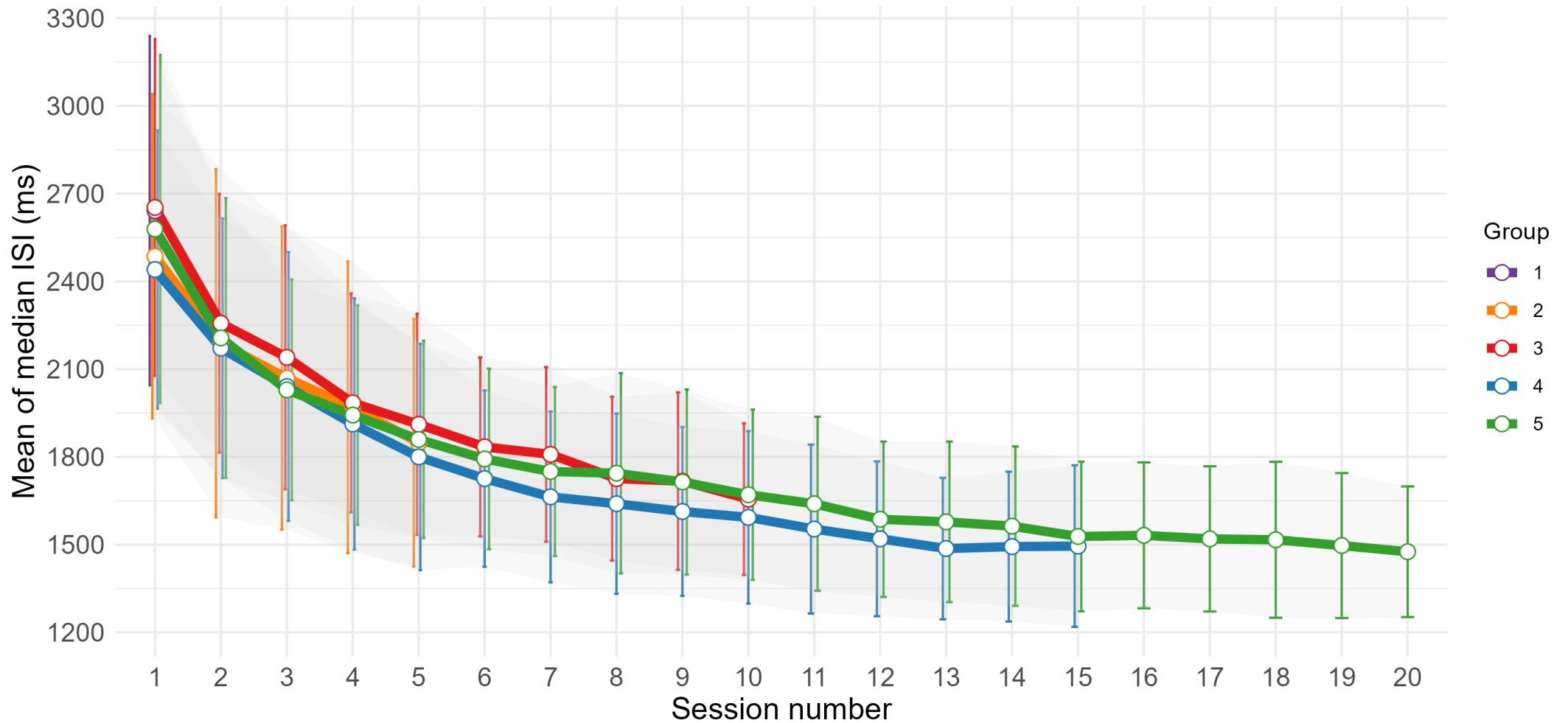


Fig. 2 Means of aPASAT median interstimulus interval per session. *Note:* Group 1 = 1 session; Group 2 = 5 sessions; Group 3 = 10 sessions; Group 4 = 15 sessions; Group 5 = 20 sessions; The waitlist control group (Group 0) is not included in this plot as these participants completed no training session.

Table 2. Mean cognitive task performance and self-report questionnaires per group over time

Variable	GR 0 baseline	GR 0 post	GR 0 FU1	GR 0 FU2	GR 1 baseline	GR 1 post	GR 1 FU1	GR 1 FU2	GR 2 baseline	GR 2 post	GR 2 FU1	GR 2 FU2
Cognitive control: PASAT (<i>SD</i>)	31.6 (10.0)	42.6 (11.0)	41.0 (11.6)	43.4 (10.6)	29.5 (13.4)	48.89 (16.7)	43.1 (17.9)	43.5 (16.5)	30.6 (19.2)	63.8 (19.5)	60.0 (18.8)	62.0 (18.8)
Cognitive control: dual n-back (<i>SD</i>)	2.8 (0.9)	3.1 (1.1)	-	-	2.7 (0.9)	3.1 (1.0)	-	-	2.6 (1.1)	3.0 (1.3)	-	-
BDI-II (<i>SD</i>)	11.3 (8.6)	12.9 (9.3)	12.7 (10.1)	11.09 (7.9)	12.7 (10.4)	12.7 (9.4)	14.4 (13.2)	13.6 (9.8)	10.1 (7.0)	9.0 (7.0)	11.7 (10.4)	11.6 (10.5)
BAT	2.4 (0.7)	2.4 (0.7)	2.3 (0.8)	2.4 (0.6)	2.4 (0.9)	2.4 (0.8)	2.4 (0.9)	2.4 (0.8)	2.2 (0.5)	2.3 (0.6)	2.3 (0.7)	2.2 (0.8)
CEQ – Credibility	-0.1 (2.7)	-0.1 (2.7)	-	-	0.0 (2.3)	-0.8 (3.0)	-	-	0.3 (1.9)	0.5 (2.2)	-	-
CEQ – Expectancy	0.0 (2.7)	-0.5 (2.9)	-	-	0.1 (3.1)	-1.2 (2.6)	-	-	0.4 (2.3)	0.0 (2.6)	-	-
CERQ – adaptive (<i>SD</i>)	58.8 (10.3)	57.9 (11.3)	56.9 (12.6)	58.3 (11.5)	61.5 (12.8)	62.2 (13.9)	56.1 (14.0)	60.3 (16.3)	62.6 (12.9)	60.2 (13.5)	58.8 (15.1)	58.8 (16.6)
CERQ – maladaptive (<i>SD</i>)	39.9 (9.6)	39.3 (9.7)	36.1 (9.5)	35.7 (8.3)	38.9 (8.9)	37.1 (8.3)	36.0 (9.8)	34.7 (9.3)	40.7 (7.7)	37.1 (6.4)	36.6 (6.0)	34.9 (8.7)
PTQ (<i>SD</i>)	35.4 (10.3)	36.0 (11.0)	34.5 (12.6)	32.4 (11.4)	35.3 (12.1)	32.7 (11.9)	31.3 (12.5)	31.1 (14.2)	33.9 (11.5)	34.0 (8.1)	34.0 (10.1)	31.4 (11.2)
RDQ (<i>SD</i>)	24.1 (16.8)	30.8 (19.4)	29.6 (19.5)	27.7 (15.4)	27.4 (17.8)	31.0 (20.2)	32.3 (23.9)	29.4 (19.4)	23.0 (14.1)	24.7 (16.4)	28.0 (18.7)	26.5 (19.0)
ATQ-EC (<i>SD</i>)	4.0 (0.9)	4.1 (0.9)	4.2 (0.8)	4.3 (0.9)	4.2 (0.9)	4.2 (0.9)	4.3 (1.1)	4.3 (1.0)	4.5 (0.8)	4.4 (0.8)	4.6 (0.8)	4.6 (0.8)
Variable	GR 3 baseline	GR 3 post	GR 3 FU1	GR 3 FU2	GR 4 baseline	GR 4 post	GR 4 FU1	GR 4 FU2	GR 5 baseline	GR 5 post	GR 5 FU1	GR 5 FU2
Cognitive control: PASAT (<i>SD</i>)	29.7 (11.8)	69.2 (12.9)	63.3 (15.8)	60.2 (18.1)	31.3 (12.8)	79.1 (12.5)	75.2 (13.5)	75.1 (12.9)	32.4 (12.6)	80.3 (12.1)	71.2 (15.5)	71.6 (15.2)
Cognitive control: dual n-back (<i>SD</i>)	2.7 (0.9)	3.2 (1.0)	-	-	2.9 (1.1)	3.3 (1.1)	-	-	2.6 (1.0)	2.9 (1.1)	-	-
BDI-II (<i>SD</i>)	13.9 (8.4)	8.9 (7.3)	12.2 (9.5)	10.7 (9.3)	12.6 (10.1)	8.8 (7.5)	8.9 (6.6)	8.7 (8.4)	9.0 (5.5)	6.1 (5.6)	8.3 (7.5)	8.0 (7.0)
BAT	2.4 (0.6)	2.3 (0.6)	2.4 (0.8)	2.4 (0.7)	2.5 (0.8)	2.2 (0.6)	2.1 (0.7)	2.2 (0.6)	2.1 (0.5)	2.0 (0.6)	2.0 (0.6)	2.1 (0.7)
CEQ – Credibility	-0.7 (2.6)	-0.4 (2.9)	-	-	0.4 (2.3)	0.5 (2.8)	-	-	0.4 (1.6)	0.2 (2.2)	-	-
CEQ – Expectancy	-0.5 (2.5)	0.4 (2.5)	-	-	-0.7 (2.3)	0.5 (2.1)	-	-	0.8 (2.0)	0.9 (2.3)	-	-
CERQ – adaptive (<i>SD</i>)	60.1 (9.9)	62.2 (11.3)	59.4 (13.7)	62.9 (12.5)	60.4 (12.8)	61.0 (14.4)	57.8 (15.7)	60.3 (15.2)	66.8 (13.0)	68.7 (13.4)	63.5 (15.3)	65.1 (16.0)
CERQ – maladaptive (<i>SD</i>)	40.0 (7.2)	38.4 (8.3)	35.2 (7.3)	35.4 (7.9)	40.4 (8.8)	37.8 (8.5)	36.6 (10.5)	35.6 (7.9)	37.0 (7.4)	33.8 (5.8)	31.5 (6.0)	32.3 (6.8)
PTQ (<i>SD</i>)	35.9 (9.5)	31.4 (11.1)	28.5 (13.1)	28.7 (12.9)	37.5 (11.3)	33.1 (10.2)	31.9 (12.7)	28.9 (12.3)	33.3 (9.6)	28.2 (10.5)	28.2 (12.2)	27.5 (11.3)

RDQ (<i>SD</i>)	28.6 (18.1)	24.1 (17.9)	28.3 (18.8)	25.6 (20.5)	24.4 (17.4)	20.4 (15.6)	25.6 (17.2)	22.7 (17.7)	21.4 (18.1)	15.4 (14.1)	18.5 (16.5)	20.1 (17.1)
ATQ-EC (<i>SD</i>)	4.2 (0.7)	4.2 (0.8)	4.2 (0.9)	4.2 (0.8)	4.2 (0.9)	4.3 (0.8)	4.4 (1.0)	4.4 (1.0)	4.4 (0.6)	4.6 (0.7)	4.6 (0.7)	4.6 (0.8)

Note: GR 0 waitlist control group, GR 1 one CCT session group, GR 2 five CCT sessions group, GR 3 ten CCT sessions group, GR 4 fifteen CCT sessions group, GR 5 twenty CCT sessions group, FU1 follow-up three months after baseline, FU2 follow-up six months after baseline, BDI-II Beck Depression Inventory II, BAT Burnout Assessment Tool, CERQ Cognitive Emotion Regulation Questionnaire, PTQ Perseverative Thinking Questionnaire, RDQ Remission from Depression Questionnaire, ATQ-EC Adult Temperament Questionnaire, Effortful Control subscale, CEQ Credibility and Expectancy Questionnaire – only measured at baseline and post, PASAT Paced Auditory Serial Addition Task, dual n-back – only measured at baseline and post.

Primary outcome

Depressive symptomatology was assessed with the BDI-II at all four time points. The linear mixed-effects model did not meet the homoscedasticity assumption, so generalized linear mixed-effects models were created, where the negative binomial regression showed better fit (AIC: 5374.9) compared to the Poisson model (AIC: 5817.2). Furthermore, the non-dispersion parameter in the negative binomial regression model was estimated to be 4.81, also indicating that this model indicates a better fit compared to a Poisson model ($\theta > 1$). Table A3 in the supplementary materials reports the results of the generalized linear mixed model, showing significant negative estimates for groups 3, 4 and 5 at post measure, reflecting a decrease in depressive symptoms. All other comparisons were non-significant. More central to the research question, and after correcting for multiplicity, post-hoc comparisons reveal only groups 3 and 5 to have a significantly lower estimate of depressive symptoms between baseline and post measure (see table 3). Hedges' g effect sizes and their standard errors were calculated for groups 3 ($g = 0.47$, $SE = 0.24$), 4 ($g = 0.47$, $SE = 0.25$) and 5 ($g = 0.87$, $SE = 0.25$), which converted into NNT resulted in group 3 $NNT = 3.82$, group 4 $NNT = 3.80$ and group 5 $NNT = 2.17$. Within-group effect size calculation using pre-post (instead of post means) Hedges' g effect sizes is reported in the supplementary materials, table A3. The left section of figure 3 displays the BDI means and standard deviations of each group over time.

An additional analysis was conducted to examine individual change in depressive symptomatology from baseline to post training, where we used the minimal clinically important difference of the BDI-II from Button et al. (2015). Within the intensive training group (i.e., all participants receiving at least 10 training sessions), 64.8% of participants showed clinical improvement (i.e., $>17.5\%$ decrease in depressive symptoms), whereas this was only the case for 50.7% and 25.7% in the limited training (1 and 5 session groups) and control groups respectively. For a visualization, see supplementary material, figure A3.

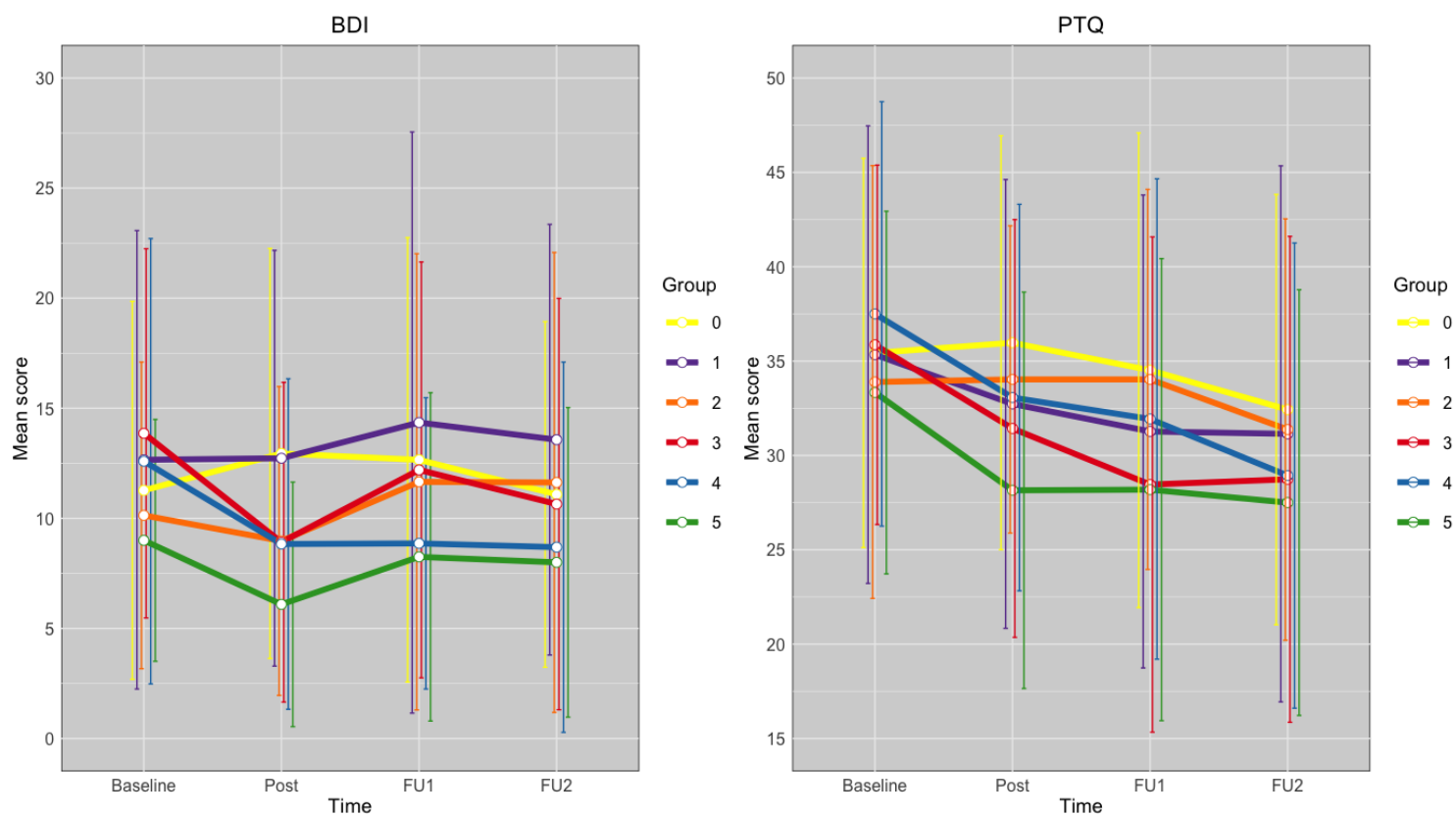


Fig. 3 Mean BDI-II and PTQ scores across groups over time

Table 3. Results of comparisons of mixed-effects model for depressive symptomatology

Comparison	Estimate	95% CI	SE	z-value	Unadjusted p-value	Adjusted p-value
Post-Pre: Group0	0.13	[-0.26, 0.53]	0.13	1.01	0.313	0.997
Post-Pre: Group1	0.00	[-0.40, 0.38]	0.13	-0.05	0.963	1.000
Post-Pre: Group2	-0.21	[-0.63, 0.20]	0.14	-1.51	0.131	0.885
Post-Pre: Group3	-0.49	[-0.90, -0.09]	0.14	-3.60	< .001*	0.006*
Post-Pre: Group4	-0.31	[-0.76, 0.13]	0.15	-2.08	0.037*	0.454
Post-Pre: Group5	-0.47	[-0.92, -0.02]	0.15	-3.10	0.002*	0.033*
FU1-Pre: Group0	0.04	[-0.36, 0.43]	0.13	0.27	0.791	1.000
FU1-Pre: Group1	0.05	[-0.34, 0.44]	0.13	0.39	0.695	1.000
FU1-Pre: Group2	0.00	[-0.41, 0.41]	0.14	0.00	0.999	1.000
FU1-Pre: Group3	-0.18	[-0.58, 0.21]	0.13	-1.38	0.167	0.940
FU1-Pre: Group4	-0.23	[-0.69, 0.23]	0.15	-1.49	0.136	0.896
FU1-Pre: Group5	-0.17	[-0.62, 0.27]	0.15	-1.16	0.248	0.988
FU2-Pre: Group0	-0.04	[-0.45, 0.36]	0.14	-0.31	0.757	1.000
FU2-Pre: Group1	0.11	[-0.28, 0.51]	0.13	0.85	0.396	1.000
FU2-Pre: Group2	0.01	[-0.40, 0.42]	0.14	0.07	0.943	1.000
FU2-Pre: Group3	-0.31	[-0.72, 0.10]	0.14	-2.26	0.024*	0.326
FU2-Pre: Group4	-0.37	[-0.83, 0.08]	0.15	-2.43	0.015*	0.220
FU2-Pre: Group5	-0.21	[-0.65, 0.24]	0.15	-1.38	0.168	0.941

Note: Group0 Waitlist control group, Group1 one CCT session group, Group2 five CCT sessions group, Group3 ten CCT sessions group, Group4 fifteen CCT sessions group, Group5 twenty CCT sessions group, Pre Baseline assessment, Post Post assessment, FU1 3 month follow-up, FU2 6 month follow-up. FWER single-step adjusted p values; * marks significance where $p < 0.05$.

Other cognitive and emotional outcome measures

Perseverative thinking as measured with the PTQ showed high variability over time, with groups 3 ($\beta = -4.55$, 95% CI [-9.02, -0.08], SE = 1.51, $z = -3.02$, $p = 0.042$) and 5 ($\beta = -4.61$, 95% CI [-9.19, -0.02], SE = 1.55, $z = -2.98$, $p = 0.048$) showing a significant decrease from baseline to post-training after correcting for multiplicity. At follow-up 1, groups 3 ($\beta = -7.52$, 95% CI [-11.99, -3.05], SE = 1.51, $z = -4.99$, $p < .001$), 4 ($\beta = -5.16$, 95% CI [-10.04, -0.29], SE = 1.64, $z = -3.14$, $p = 0.028$) and 5 ($\beta = -5.16$, 95% CI [-9.80, -0.52], SE = 1.56, $z = -3.30$, $p = 0.017$) also showed significant decreases in the tendency of individuals to engage in repetitive negative thinking patterns. These effects appeared to be maintained at follow-up 2 (group 1: $\beta = -4.67$, 95% CI [-9.10, -0.24], SE = 1.49, $z = -3.13$, $p = 0.030$), group 3 ($\beta = -7.25$, 95% CI [-11.76, -2.74], SE = 1.52, $z = -4.77$, $p < .001$), group 4 ($\beta = -8.16$, 95% CI [-13.04, -3.29], SE = 1.64, $z = -4.97$, $p < .001$) and group 5 ($\beta = -5.85$, 95% CI [-10.48, -1.21], SE = 1.56, $z = -3.74$, $p = 0.003$)). The right section of figure 3 displays the PTQ means and standard deviations of each group over time.

The adaptive compound scores for the CERQ demonstrated no significant differences between groups in *adaptive emotion regulation strategy use* at post assessment. A significant negative estimate emerged for group 1 at follow-up 1 ($\beta = -5.38$, 95% CI [-10.72, -0.05], SE = 1.80, $z = -2.99$, $p = 0.046$), but was no longer significant at follow-up 2 ($\beta = -1.98$, 95% CI [-7.42, 3.45], SE = 1.83, $z = -1.08$, $p = 0.993$). R^2_C was 0.69 for the model. After adjusting p values for multiplicity, we observed greater variability in *maladaptive emotion regulation strategy use*, with a significant decrease for group 2 at post assessment ($\beta = -3.87$, 95% CI [-7.31, -0.44], SE = 1.16, $z = -3.34$, $p = 0.014$) and significant differences at follow-up 1 (group 0: $\beta = -4.17$, 95% CI [-7.60, -0.74], SE = 1.16, $z = -3.61$, $p = .005$; group 2: $\beta = -4.39$, 95% CI [-7.83, -0.95], SE = 1.16, $z = -3.79$, $p = .003$; group 3: $\beta = -4.62$, 95% CI [-8.07, -1.18], SE = 1.16, $z = -3.98$, $p = .001$; group 5: $\beta = -5.63$, 95% CI [-9.20, -2.06], SE = 1.20, $z = -4.68$, $p < .001$). At follow-up 2, all groups showed significant decreases (group 0: $\beta = -4.54$, 95% CI [-7.97, -1.11], SE = 1.16, $z = -3.93$, $p = .002$; group 1: $\beta = -4.52$, 95% CI [-7.93, -0.97], SE = 1.15, $z = -3.93$, $p = .001$; group 2: $\beta = -6.10$, 95% CI [-9.54, -2.67], SE = 1.16, $z = -5.27$, $p < .001$; group 3: $\beta = -4.40$, 95% CI [-7.88, -0.92], SE = 1.17, $z = -3.76$, $p = .003$; group 4: $\beta = -4.20$, 95% CI [-7.96, -0.45], SE = 1.27, $z = -3.32$, $p = .015$; group 5: $\beta = -4.82$, 95% CI [-8.39, -1.25], SE = 1.20, $z = -4.01$, $p = .001$). The R^2_C was 0.68 for the model.

In line with the BDI-II, the RDQ model suggested an immediate reduction in level of *residual complaints* for groups 3, 4 and 5. However, these effects were not significant following

adjustment for multiple comparisons (all p 's > 0.695). Similar results were obtained for both follow-ups, where no significant differences emerged. The variance explained by the model (R_C^2) was 0.55.

After correcting for multiplicity, the linear mixed model of the BAT showed no significant differences between groups at post assessment in *burnout complaints*. On follow-up 1, a significant difference for group 4 ($\beta = -0.32$, 95% CI [-0.62, -0.02], SE = 0.09, $z = -3.17$, $p = 0.026$) emerged, which was no longer significant at follow-up 2 ($\beta = -0.28$, 95% CI [-0.57, 0.02], SE = 0.10, $z = -2.74$, $p = 0.098$). All other comparisons for the BAT did not yield significant results. The variance explained by the model (R_C^2) was 0.70.

Similar to the dual n-back, the CEQ was only administered at baseline and post. First, the *credibility* subscale was analyzed. Likelihood ratio tests were conducted to compare the fit of two linear mixed-effects models, the first including only the main effects of group and time, and the second also containing their interaction. Here, the second model (with the interaction term) did not show an improvement in fit ($\chi^2(6) = 4.88$, $p = 0.430$; AIC model 1: 1927.5, AIC model 2: 1932.6), so we examined the first model in our analyses. No main effect of group (all p 's > 0.283) or time ($\beta = -0.08$, SE = 0.18, $t(210.40) = -0.43$, $p = 0.669$) was found. The conditional R^2 variance explained by this model was 0.43. Second, the *expectancy* subscale was analyzed, using the same analytical procedure. Here, the model with the interaction between group and time showed better fit ($\chi^2(6) = 24.6$, $p < .001$; AIC model 1: 1942.3, AIC model 2: 1927.7). The second model found a significant comparison for group 3 ($\beta = 1.38$, SE = 0.58, $t(204.74) = 2.37$, $p = 0.019$) and for group 4 ($\beta = 1.65$, SE = 0.60, $t(206.17) = 2.74$, $p = 0.007$), but not for the other groups (all p 's > 0.13). The random effects of the model indicated significant variance at the participant level ($\sigma^2 = 3.36$) and within-participant residuals ($\sigma^2 = 3.01$), suggesting variability both between participants and within individuals across time points. Conditional R^2 was 0.55.

Discussion

This RCT examined the dose-response relationship of an aPASAT training on depressive symptomatology in individuals with remitted depression. The key findings of the current study is that groups receiving at least 10 training sessions showed less depressive symptoms and decreased perseverative thinking shortly after training. At follow-up, only perseverative thinking remained decreased for groups with a minimum of 10 training sessions. No group differences emerged on broader indices of the use of (mal)adaptive emotion regulation strategies. We discuss the main findings in more detail below.

In line with most CCT research, we found task-specific transfer: PASAT accuracy scores improved over time in all groups. Additionally, the linear mixed model showed a significant group by time interaction effect, suggesting that groups who completed higher amounts of aPASAT training sessions showed stronger task-specific transfer over time compared to lower intensity groups. The dual n-back was used to assess near transfer, where we found a significant main effect of time, indicating that all groups showed stronger performance at post measure compared to baseline. Interestingly, there was no significant interaction effect, indicating that all groups improved similarly. The absence of near transfer is in line with prior CCT research, which has inconsistently found evidence for near transfer effects (Soveri et al., 2017). Possible reasons for not finding near transfer could be that working memory did not improve sufficiently to be detected in an untrained cognitive task. Another possibility is that the dual n-back was not receptive enough to potentially detect small cognitive gains due to its limited range in difficulty level. This becomes clear when we compare this to another cognitive task. For example, the blocks of the standardized PASAT can be varied in terms of timing to match the cognitive capability of the sample by changing the speed of the blocks. In samples with significantly reduced cognitive capability, the stimulus presentation in the timed blocks can be adjusted to account for this decreased cognitive capability. For example, Martin et al. (1996) used the PASAT in blocks of 4s and 2s in patients with sleep apnea. When testing a cognitively stronger sample, faster blocks can be used. For example, Diehr et al. (1998) used a 200-item version of the PASAT with blocks of 3s, 2.4s, 2s and 1.6s. In comparison, the dual n-back is limited in its difficulty range and does not allow for precision adjustments. If the $n = 1$ level is too easy and the $n = 3$ level too hard for almost everyone, this only allows for limited variability in the $n = 2$ level, which may not be sufficient to detect small cognitive gains. Future research should consider using cognitive tasks with a wider range and increased sensitivity in order to be able to detect potentially small near-transfer effects.

The effortful control subscale of the ATQ, which was used to assess subjective cognitive control, showed no significant changes following CCT in any of the groups. This indicates that, despite improved performance on the standardized PASAT, there were no differences in subjective experiences of cognitive control. Indeed, subjective and objective measures of cognitive functioning are often weakly correlated (Hagen et al., 2021), although the reasons for this discrepancy are currently unclear.

Generalized mixed effects models found reductions in depressive symptoms immediately after training in groups receiving at least 10 training sessions. No differences emerged with groups receiving five or less sessions immediately after training. At three and six months after baseline, negative estimates (indicating decreased depressive symptoms) remained for the higher intensity training groups but did not reach significance. This is inconsistent with previous studies that have demonstrated long-term beneficial effects of CCT (Siegle et al., 2014, Hoorelbeke et al., 2021). It is possible that repeated administration over time is required in order to maintain training-related gains, as suggested by Hoorelbeke and colleagues (2022). The medium to large effect sizes for the training groups receiving at least 10 training sessions are slightly higher than the previously reported small to medium effect sizes for aPASAT training in a recent meta-analysis (Vander Zwalmen et al., 2023), although it is important to note that these included depressed, remitted depressed, at-risk and healthy samples, whereas this study included only individuals with remitted depression. Indeed, it could be possible that larger effect sizes of CCT would be found in samples with more residual cognitive symptoms.

Interestingly, and in line with previous research, perseverative thinking was found to be sensitive to CCT. At post-measurement, groups 3 and 5 showed a significant decrease in perseverative thinking. Similar to the depressive symptoms data, group 4 also showed a decrease in the model, but did not reach significance after controlling for multiple testing. Especially noteworthy is that at follow-up, when the depressive symptoms no longer showed significant changes, perseverative thinking continued to decrease for groups 3, 4 and 5, at both follow-ups. This is in line with previous studies, where perseverative thinking and rumination have also been shown to be receptive to CCT (Hoorelbeke et al., 2015, Siegle et al., 2007), and sustained emotional transfer effects have been reported following 10 sessions of aPASAT training (Hoorelbeke & Koster, 2017; Hoorelbeke et al., 2022).

Previous research has suggested that changes in the use of emotion regulation strategies could be one of the potential mechanisms through which CCT might affect depressive symptoms. Two possible ways through which CCT could have a beneficial effect on depression

are by either increasing the use of adaptive-, or by decreasing the use of maladaptive emotion regulation strategies. Indeed, theories investigating the working mechanisms of CCT often focus on the role of maladaptive emotion regulation strategies, including forms of perseverative thinking or rumination (Koster et al., 2011, Watkins & Roberts, 2020). Although we observed beneficial effects of CCT on perseverative thinking, this did not transfer to broader indices of maladaptive emotion regulation strategy use. That is, the use of maladaptive emotion regulation strategies significantly decreased over time, albeit unrelated to training dosage. In this context, it is particularly interesting that long-term effects were found only for perseverative thinking and not for depressive symptoms. Moreover, in line with prior aPASAT studies in RMDs, we found no meaningful effects of training on the use of adaptive emotion regulation strategies (e.g., Hoorelbeke et al., 2023). Taken together, further research into the temporal unfolding of CCT effects, and the role of emotion regulation in particular, is required to better understand the specific mechanisms responsible for training related improvements.

As a more exploratory measure, given its close relation with depression, we also assessed burn-out symptoms. However, we found no evidence that CCT affected burn-out symptoms in any of the groups. Moreover, the RDQ, providing a broader estimate of functioning, also showed no significant changes between groups or over time. One possible reason for this is that the RDQ, in addition to the two subscales for depressive symptoms, also has several subscales that focus on positive aspects of life, whereas the BDI-II only focusses on depressive symptoms.

Importantly, the credibility subscale of the CEQ did not show differences in our analysis, indicating that participants rated their belief in the intervention similarly, regardless of group. The expectancy subscale showed a significant interaction effect between group and time for groups 3 and 4, indicating a higher expectation of improvement following CCT. Group 5 scored slightly higher on baseline expectancy and did not show a significant interaction effect. It is not surprising that the groups receiving no or low doses of CCT did not expect differences in the effectiveness of the intervention, whereas the higher expectancy scores might be related to the decreased depressive symptoms in these same groups. It is important to note that for the CEQ data both within-participant and between-participant variance was substantial, highlighting the importance of interpersonal differences. Indeed, as with other antidepressant interventions, personal preferences may play an important role in the expectancy of an intervention.

Examination of dosage is of major importance to consider in potential clinical use. In addition to the total number of training sessions, other variables may have a similar impact on CCT dosing and efficacy. For example, based on input of RMD individuals (Vervaeke et al., 2018), most of the recent gamified aPASAT studies – including this one – used training sessions of 15 minutes with a regimen of one per day (Vander Zwalm et al., 2023), but other CCT studies have used different paradigms, including shorter or longer sessions, or fewer sessions per week, but more weeks of training. Recent studies with an n-back working memory training have generally used slightly longer training sessions of 20 to 30 minutes and a higher dosage, around 20 sessions (e.g., Beloe & Derakshan, 2019; Zhang et al., 2019). Similar to the high dosage groups in this study, both Beloe and Derakshan (2019) and Zhang et al. (2019) reported reduced depressive symptomatology following CCT. Another way to investigate CCT duration could be to examine total session duration in minutes, rather than the total number of sessions. This is one way in which CCT could potentially be personalized in a clinical implementation, taking into account an individual's availability. Other strategies to examine effects of dosage is through administering additional CCT sessions (i.e., booster sessions) at times when individuals experience increased symptoms (Hoorelbeke et al., 2022). In this study, we did not observe any floor effects for depressive symptoms, suggesting that further improvements on indicators of emotional and cognitive functioning may be possible in higher doses or other CCT implementations. Further research could examine whether different training configurations or the addition of booster sessions could lead to stronger improvements or longer-lasting training-induced effects.

Conclusion

In recent years, CCT has emerged as a promising intervention in the context of depression. This study, in a sample of individuals with remitted depression, suggests that at least 10 training sessions are required to find reductions in depressive symptoms shortly after training. No long-term effects on depressive symptoms were found, suggesting the need for continued training or the use of booster sessions in order to maintain training-induced effects. Perseverative thinking was shown to be susceptible to CCT both shortly after training, as well as long-term, but only for the groups that completed at least 10 training sessions.

Author Contributions – CRediT

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Conflicts of Interest

The authors declare that there are no conflicts of interest.

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Supplemental Material

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Supplementary material

Name of software package	Version number
R	4.3.2
openxlsx	4.2.5.2
lme4	1.1-35.1
emmeans	1.9.0
lmerTest	3.1-3
tidyverse	2.0.0
patchwork	1.2.0
reshape2	1.4.4
multcomp	1.4-25
effectsize	0.8.6
MuMIn	1.47.5
esc	0.5.1
dmetar	0.1.0

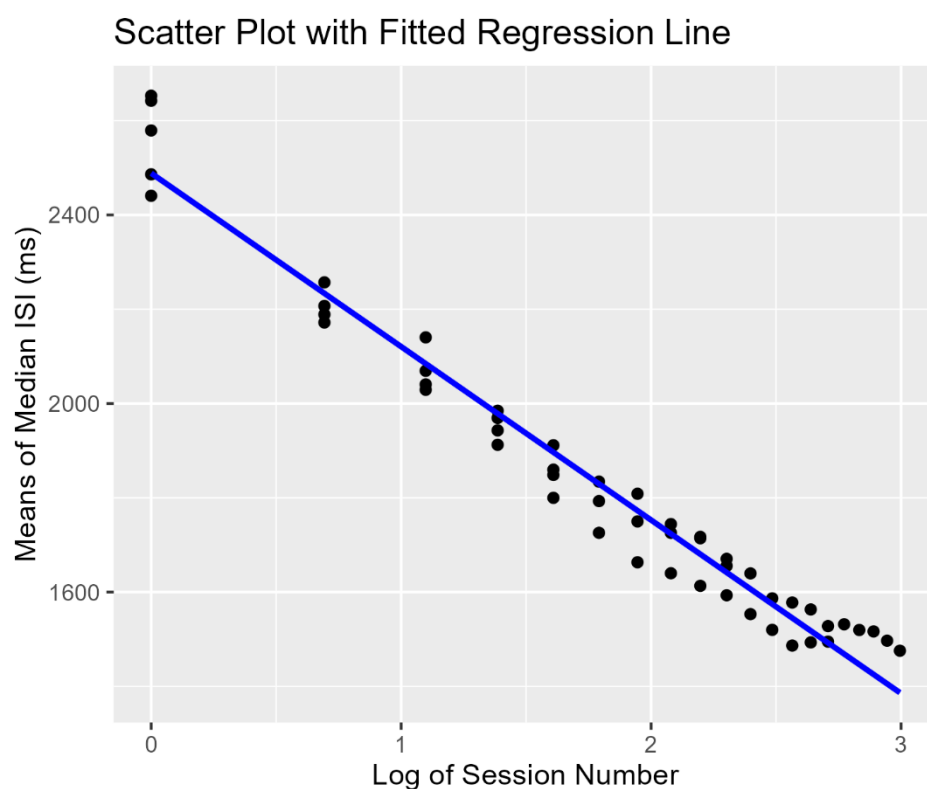
Table A1 Summary of software packages and version numbers**Fig. A1** Training progression: means of median Interstimulus Interval (ISI) in function of natural logarithm of session number

Table A2 Pre-post BDI correlations, pre-post intervention Hedges' g effect sizes, their standard errors and NNT/NNH.

Group	Correlations	Hedges' g	SE	Positive number = Number Needed to Treat (NNT), Negative number = Number Needed to Harm (NNH)
Group 0	r = 0.666	g = -0.14	0.13	-12.7
Group 1	r = 0.771	g = 0.00	0.11	/
Group 2	r = 0.612	g = 0.14	0.15	12.7
Group 3	r = 0.679	g = 0.50	0.15	3.62
Group 4	r = 0.820	g = 0.25	0.11	7.13
Group 5	r = 0.690	g = 0.40	0.14	4.49

Table A3 Results from multiple comparisons of the PASAT linear mixed model

Comparison	Estimate	95% CI	SE	z-value	Adjusted p-value
Post-Pre: Group0	0.11	[0.06, 0.16]	0.02	6.29	< .001*
Post-Pre: Group1	0.19	[0.14, 0.24]	0.02	11.44	< .001*
Post-Pre: Group2	0.31	[0.26, 0.37]	0.02	18.04	< .001*
Post-Pre: Group3	0.39	[0.34, 0.44]	0.02	22.83	< .001*
Post-Pre: Group4	0.47	[0.42, 0.53]	0.02	25.70	< .001*
Post-Pre: Group5	0.49	[0.43, 0.54]	0.02	27.36	< .001*
FU1-Pre: Group0	0.09	[0.04, 0.14]	0.02	5.18	< .001*
FU1-Pre: Group1	0.13	[0.08, 0.18]	0.02	7.69	< .001*
FU1-Pre: Group2	0.28	[0.23, 0.33]	0.02	15.87	< .001*
FU1-Pre: Group3	0.33	[0.28, 0.38]	0.02	19.28	< .001*
FU1-Pre: Group4	0.43	[0.37, 0.48]	0.02	22.90	< .001*
FU1-Pre: Group5	0.40	[0.34, 0.45]	0.02	22.01	< .001*
FU2-Pre: Group0	0.11	[0.06, 0.16]	0.02	6.60	< .001*
FU2-Pre: Group1	0.14	[0.09, 0.19]	0.02	7.97	< .001*
FU2-Pre: Group2	0.30	[0.24, 0.35]	0.02	17.00	< .001*
FU2-Pre: Group3	0.30	[0.25, 0.35]	0.02	17.17	< .001*
FU2-Pre: Group4	0.43	[0.37, 0.48]	0.02	22.82	< .001*
FU2-Pre: Group5	0.40	[0.35, 0.45]	0.02	22.27	< .001*

Note: Group0 Waitlist control group, *Group1* one CCT session group, *Group2* five CCT sessions group, *Group3* ten CCT sessions group, *Group4* fifteen CCT sessions group, *Group5* twenty CCT sessions group, *Pre* Baseline assessment, *Post* Post assessment, *FU1* 3 month follow-up, *FU2* 6 month follow-up. FWER single-step adjusted p values with significance level:

* p < 0.05.

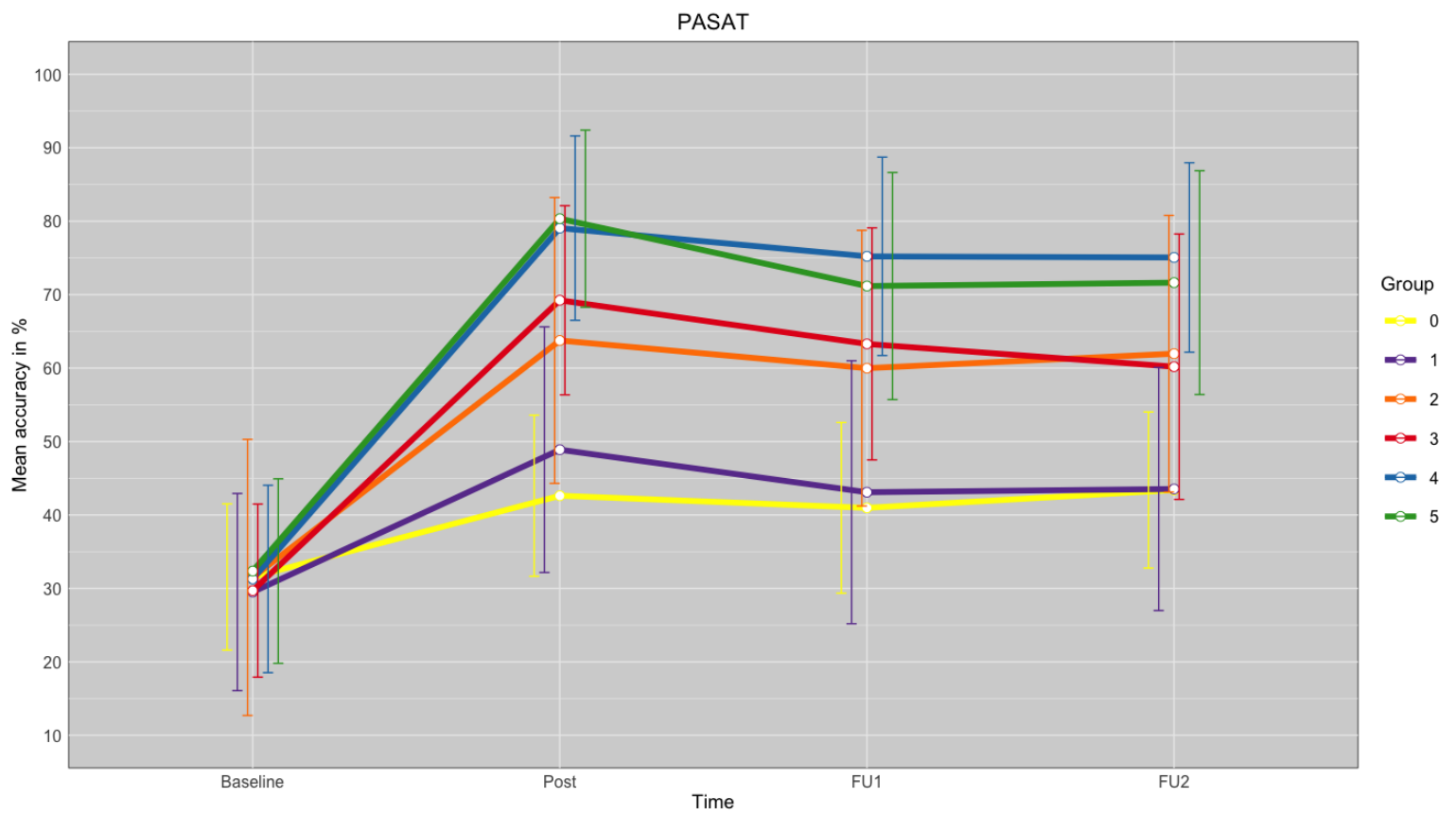


Fig. A2 PASAT performance per group over time

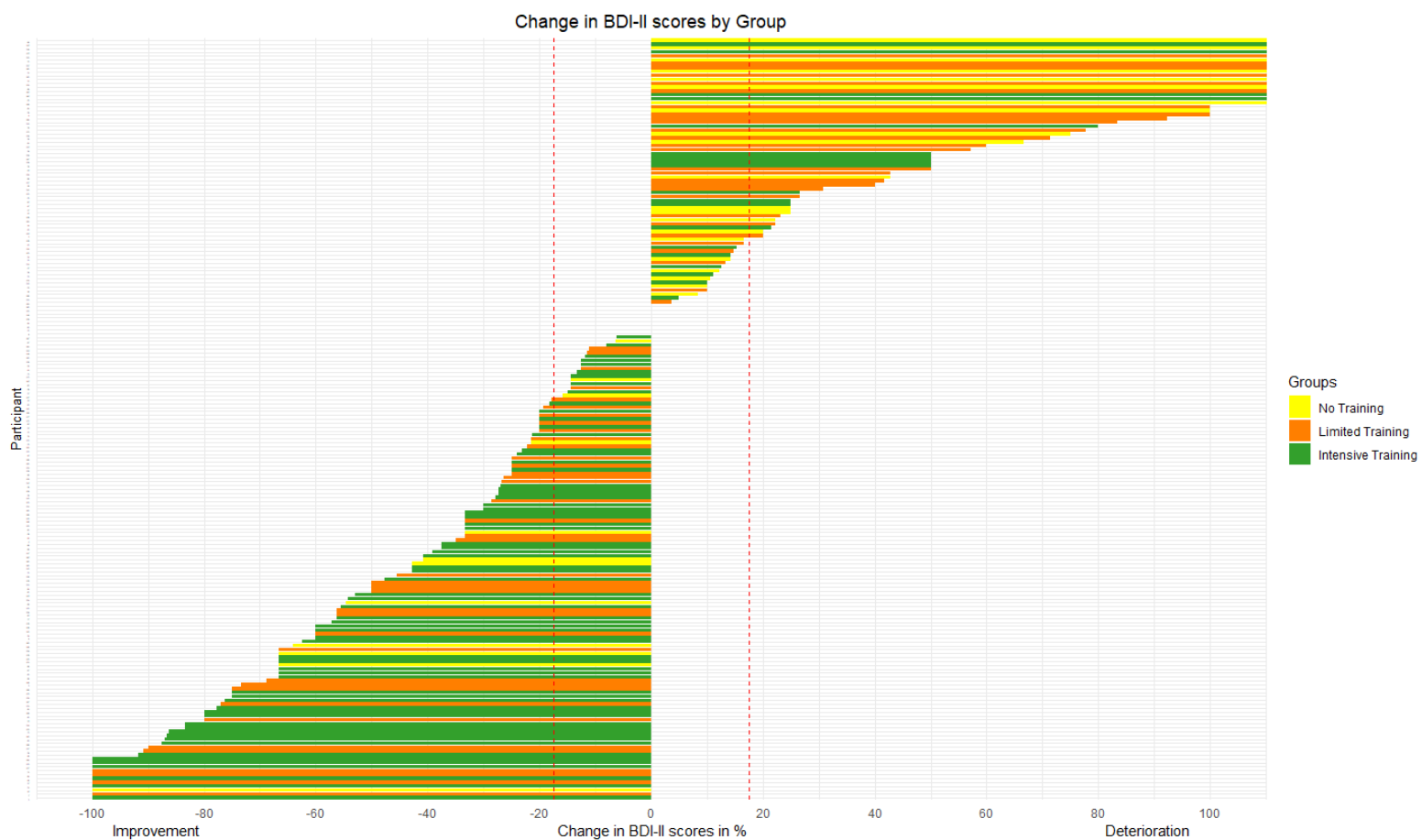


Fig. A3 Change in BDI-II (post to baseline) in percentage

Note: No training: waitlist control group. Limited training: Groups 1 (1 session) and 2 (5 sessions). Intensive training: Groups 3 (10 sessions), 4 (15 sessions) and 5 (20 sessions).